

Department of Computer Engineering

HiTec University, Taxila

Assignment # 1

Digital Image Processing (CS-406)



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Assignment # 01

Q#1 What are the application of image segmentation in the Smart city?
Define any five applications. (Note: should not exceed more than 15 lines)

ANS:IMAGE SEGMENTATION:

IMAGE SEGMENTATION IS A CRUCIAL COMPUTER VISION TECHNIQUE THAT DIVIDES IMAGES INTO REGIONS FOR DETAILED ANALYSIS,AIDING IN THE MANAGEMENT, MONITORING AND AUTOMATION OF SERVICES IN SMARTCITIES.THE APPLICATION OF IMAGE SEGMENTATION FOLLOWING ARE:

1:TRAFFIC MANAGEMENT

2:PUBLIC SAFETY AND SURVEILLANCE

3:WASTE MANAGEMENT

4:INFRASTRUCTURE MANAGEMENT

5:GREEN SPACE MANAGEMENT

1.TRAFFIC MANAGEMENT:

IMAGE SEGMENTATION AIDS IN REAL-TIME TRAFFIC FLOW MONITORING,ACCIDENT DETECTION AND SIGNAL OPTIMIZATION TO REDUCE CONGESTION BY IDENTIFYING AND TRACKING VEHICLES,PEDESTRIANS AND ROAD INFRASTRUCTURE.

2.PUBLIC SAFETY AND SURVEILLANCE:

SMART SURVEILLANCE SYSTEMS USE VIDEO FOOTAGE SEGMENTATION TO DETECT UNUSUAL BEHAVIORS OR INTRUSIONS,IMPROVING URBAN SECURITY AND FACILITATING SWIFT EMERGENCY RESPONSES.

3.WASTE MANAGEMENT:

SEGMENTATION IS SMART BINS AIDS IN AUTOMATED WASTE SORTING FOR RECYCLING,REDUCING MANUAL LABOR AND CONTRIBUTING TO MORE EFFICIENT AND ECO-FRIENDLY WASTE MANAGEMENT SYSTEMS.

4.INFRASTRUCTURE MONITORING:

IMAGE SEGMENTATION USE AERIAL OR STREET-LEVEL IMAGERY TO DETECT STRUCTURAL DAMAGE IN BRIDGES,ROADS AND BUILDINGS,ENHANCING THEIR LONGEVITY AND SAFETY THROUGH PREVENTIVE MAINTENANCE.

5.GREEN SPACE MANAGEMENT:

IMAGE SEGMENTATION AIDS IN IDENTIFYING URBAN STRUCTURES AND VEGETATION,ENABLING CITY MONITORING AND MAINTENANCE OF PARKS,GARDENS AND

GREEN ROOFS,THUS PROMOTING SUSTAINABILITY AND ENVIRONMENTAL BALANCE.

Q#2 Develop a Matlab code to zoom image using pixel replication by the factor of 2 and more. (Note: code to zooming by factor 2 has given in the lecture#3)

MATLAB CODE:

```
% Read the image
```

```
img = imread('image.jpg'); % Replace with your image path
```

```
figure, imshow(img), title('Original Image');
```

```
% Define zoom factor (you can change this factor to zoom more)
```

```
zoom_factor = 3; % Set the zoom factor (e.g., 2, 3, etc.)
```

```
% Get the size of the original image
```

```
[rows, cols, channels] = size(img);
```

```
% Preallocate the zoomed image with the new size
```

```
zoomed_img = zeros(rows * zoom_factor, cols * zoom_factor, channels,  
'uint8');
```

```
% Perform pixel replication
```

```
for i = 1:rows
```

```
    for j = 1:cols
```

```

% Replicate each pixel by zoom_factor

zoomed_img((i-1)*zoom_factor+1:i*zoom_factor, (j-
1)*zoom_factor+1:j*zoom_factor, :) = repmat(img(i, j, :), [zoom_factor,
zoom_factor, 1]);

end

end

% Display the zoomed image

figure, imshow(zoomed_img), title(['Zoomed Image by a factor of ',
num2str(zoom_factor)]);

```

OUTPUT:

